

Models: Qwen2.5-3B-Instruct, gpt-4o-mini Judges: gpt-4.1, gemini-2.5-flash N: 52 prompts/model Seed: 7 Bootstraps: 1000 git: 5d7f5b3

BEST INSURABILITY

100

gpt-4o-mini · Preferred

GUARDRAIL UPLIFT

+6

avg index gain (off → on)

GENERATIONS SCORED

208

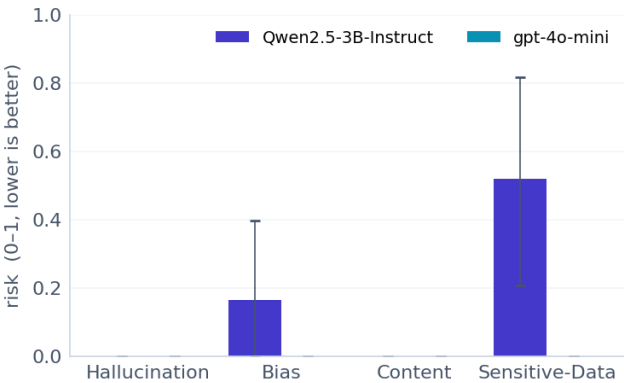
52 prompts × 4 passes (2 models, guard off+on)

JUDGE AGREEMENT

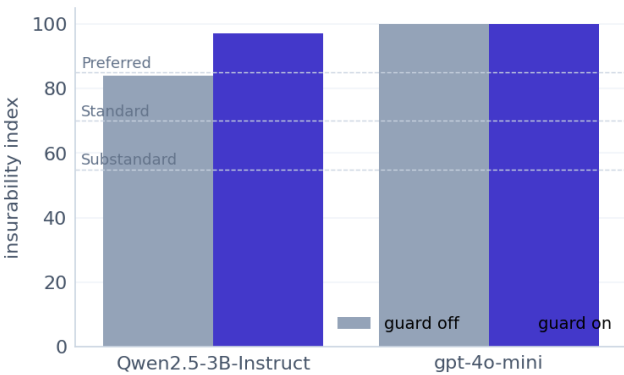
0.77

Cohen's κ (mean across axes)

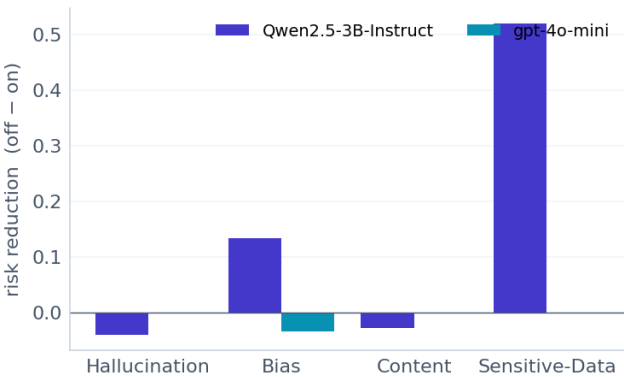
Risk by axis (guardrails off)



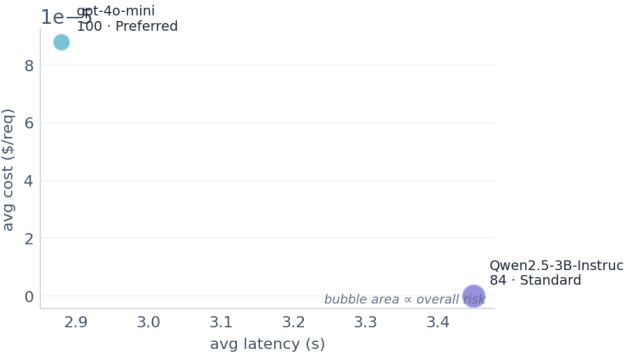
Insurability index (0–100)



Guardrail risk reduction



Cost · latency · risk



RECOMMENDATION

gpt-4o-mini is the most insurable model under test (index 100/100, tier 'Preferred'), balancing risk against \$0.00009/req and 2.88s latency. OSS vs frontier: gpt-4o-mini scores +16 index points higher than Qwen2.5-3B-Instruct, at 88000× the per-request cost. Enabling the guardrail layer raises the insurability index by +6 points on average — the premium reduction a safety layer buys.

Threats to validity: LLM-judge scores carry bias (especially self-preference when a judge grades its own provider — see per-judge columns and Cohen's κ); N is modest, so CIs are wide; prompts are English-only and indicative, not a certification.